Song, Ge

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Education

2021 – 2025 **Ph.D., University of South Carolina** Mechanical Engineering
Dissertation title: *Recognizing the Unexpected: Deep Learning Across Complex Environments*

2019 – 2021 **M.S., Boston University** Mechanical Engineering
Thesis title: *Comparison of Tracking Algorithm for Differential Drive Robots*

2015 – 2019 **B.S., Nanjing University of Science and Technology** Mechanical Engineering
Thesis title: *Design of Automatic Sorting System for Parts*

Professional Appointments

2026 – present **Postdoctoral Research Associate, University of Connecticut** Mechanical Engineering
Research focus: *Microstructure Characterization and Reconstruction*
Process-Property Modeling
Additive Manufacturing Design and Automation
Large Language Model Knowledge Retrieval

Research Experience

- **Discover Influential Geosocial Factors Aggravating Crossing Trespassing**.

  1. Developed an AI-enabled framework for evaluating the risk of trespassing incidents at railroad crossings, integrating geographic, demographic, and behavioral data.
  2. Conducted advanced geospatial and demographic data analysis, utilizing mapping tools and Census datasets to identify potential trespassing 'hotspots'.
  3. Led collaboration with an industrial partner, TRAINFO, to analyze proprietary long-term monitoring datasets including trespassing volume, land use around crossings, and paths connecting crossings, and validate predictive models in real-world scenarios.

- **Real-Time Health and Security Monitoring and Diagnosis of Manufacturing Systems Based on Energy Consumption Auditing**.

  1. Developed a side-channel energy consumption auditing system on a Raspberry Pi for data logging during robot operation
  2. Integrated OctoPrint and ROS on a Raspberry Pi to monitor the operating status of a 3D printer
  3. Deployed the model on the edge computing device, Jetson AGX Orin, achieving real-time health monitoring and anomaly detection

- **Intelligent Abnormal Situation Awareness Platform**.

  1. Developed a real-time deep-learning-based anomaly detection system, including a low-cost camera and an edge computing device, to monitor pedestrians' behaviors at grade crossings and localize and alert pedestrians with abnormal motion patterns
  2. Adapted pose estimation model OpenPose to numerically represent behaviors as coordinate sequences of pedestrian's skeleton joints in real time.

- **Generative AI for Microstructure-Property Prediction of Battery Electrodes**.

  1. Developed a generative AI framework to synthesize realistic 3D battery-electrode microstructure from 2D reference image, addressing the high cost of acquiring 3D battery microstructures.
  2. Validated generated structures against physically grounded statistical descriptors, principally the two-point correlation function (S_2) along with phase fraction, specific surface area, and percolation/tortuosity, to ensure transport-relevant fidelity rather than mere visual plausibility.
  3. Augmented the scarce experimental dataset with generated microstructures to enable downstream transport simulation and effective-property prediction at a scale infeasible with real data alone.

Research Publications

Journal Articles

1. J. Ou, **G. Song**, J. Guo, Y. Cao, and Y. Wang, "Gpu-enabled decentralized, multi-robot path planning based on global evolutionary dynamic programming and local particle swarm optimization," *Expert Systems with Applications*, vol. 321, p. 132 321, 2026.
2. J. Guo, **G. Song**, and Y. Wang, "A memory and retrieval transformer-based unsupervised learning model for anomaly detection and segmentation," *Pattern Recognition*, p. 113 004, 2025.
3. J. Ou, **G. Song**, and Y. Wang, "Graphics processing unit-enabled path planning based on global evolutionary dynamic programming and local genetic algorithm optimization," *Applied Soft Computing*, p. 113 167, 2025.
4. **G. Song**, J. Ou, Y. Cao, and Y. Wang, "An energy auditing system based on learnable positional embedding and edge computing for in-situ monitoring of 3d printing processes," *The International Journal of Advanced Manufacturing Technology*, pp. 1–17, 2025.
5. **G. Song**, Y. Qian, and Y. Wang, "Stgcn-pad: A spatial-temporal graph convolutional network for detecting abnormal pedestrian motion patterns at grade crossings," *Pattern Analysis and Applications*, vol. 28, no. 1, pp. 1–17, 2025.
6. **G. Song**, S. H. Hong, T. Kyzer, and Y. Wang, "U-TFF: A u-net-based anomaly detection framework for robotic manipulator energy consumption auditing using fast fourier transform," *Applied Sciences*, vol. 14, no. 14, p. 6202, 2024.
7. X. Wei et al., "Narrowing signal distribution by adamantane derivatization for amino acid identification using an α-hemolysin nanopore," *Nano Letters*, vol. 24, no. 5, pp. 1494–1501, 2024.
8. J. Ou, S. H. Hong, **G. Song**, and Y. Wang, "Hybrid path planning based on adaptive visibility graph initialization and edge computing for mobile robots," *Engineering Applications of Artificial Intelligence*, vol. 126, p. 107 110, 2023.
9. **G. Song**, S. H. Hong, T. Kyzer, and Y. Wang, "Energy consumption auditing based on a generative adversarial network for anomaly detection of robotic manipulators," *Future Generation Computer Systems*, vol. 149, pp. 376–389, 2023.
10. **G. Song**, Y. Qian, and Y. Wang, "Analysis of abnormal pedestrian behaviors at grade crossings based on semi-supervised generative adversarial networks," *Applied Intelligence*, vol. 53, no. 19, pp. 21 676–21 691, 2023.
11. Z. Jiang, **G. Song**, Y. Qian, and Y. Wang, "A deep learning framework for detecting and localizing abnormal pedestrian behaviors at grade crossings," *Neural Computing and Applications*, vol. 34, no. 24, pp. 22 099–22 113, 2022.

Conference Proceedings

1. J. Guo et al., "A dual-contrastive-attention transformer for unsupervised anomaly detection in lamb waves structural health monitoring," in *Annual Conference of the PHM Society*, vol. 17, 2025.
2. **G. Song**, S. H. Hong, T. Kyzer, and Y. Wang, "An energy consumption auditing anomaly detection system of robotic manipulators based on a generative adversarial network," in *Annual Conference of the PHM Society*, vol. 15, 2023.
3. **G. Song**, Y. Qian, and Y. Wang, "A deep generative adversarial network (GAN)-enabled abnormal pedestrian behavior detection at grade crossings," in *SoutheastCon 2023*, IEEE, 2023, pp. 677–684.

Skills

Languages	English, Mandarin Chinese, French.
Software	Python, MATLAB, ROS, C++, PLC, …
Hardware	Hands-on experience with Jetson AGX Xavier, Jetson AGX Orin, Raspberry Pi, 3D printer, energy sensors, mobile robot, unmanned aerial vehicle, and robotic arm.
Misc.	Team-spirited, diligent, responsible, open-minded, and self-motivated.

Awards and Achievements

2023	**Best Paper Nomination**, Annual Conference of the PHM society.
	Third Place in Graduate Student Paper Competition, IEEE Southeast Conference 2023.
2026	**Invited Poster Presenter**, MIT Manufacturing Week 2026.